Name: ____________________

Worksheet

Calculating any term

Task A: Generating sequences using n^{th} terms

1) Use substitution to find the first five terms of these arithmetic sequences.

a) $5n + 3$

Term number (n)	1	2	3	4	5
Term					

b) $0.05n + 0.3$

Term number (n)	1	2	3	4	5
Term					

c) $3 - 0.7n$

Term number (n)	1	2	3	4	5
Term					

d) $-3 - 0.7n$

Term number (n)	1	2	3	4	5
Term					

2) Jun says:

The sequence $n + 3$ must go up in three's because it says, "+3"

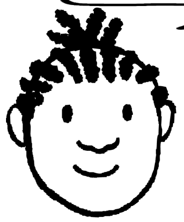
Use substitution to find the first five terms and show Jun that he is mistaken.

Term number (n)	1	2	3	4	5
Term					

Task B: Using n^{th} terms to find any term

1) The first five terms of the arithmetic sequence $42 - 5n$ are 37, 32, 27, 22, and 17.

Jacob wants to find the 20th term.



Spot Jacob's error and explain a better method to him.

2) Find the 20th, 100th and 1000th terms of the below sequences

a) $7n - 6$

b) $6 - 7n$

c) $0.06 - 0.7n$

d) $n^2 + 3n$

e) $3n + 15 - 4n^2$

Task C: Comparing n^{th} terms of different sequences

Find the first five terms of each of these sequences and compare them:

a) $3n$

b) $3n + 2$

c) $3n - 1$

d) $3n + 0.7$

e) $3n + \frac{1}{7}$

f) $-3n + 7$

g) $2 - 3n$

h) n^3